

LAB EXPERIMENTS

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33. Write a Python program to draw a scatter plot with empty circles taking a random distribution in X and Y and plotted against each other.

Code :

```
import numpy as np
import matplotlib.pyplot as plt

# Generate random values for X and Y
x = np.random.randn(50)
y = np.random.randn(50)

# Create scatter plot with empty circles
plt.scatter(x, y, facecolors='none', edgecolors='green', marker='o')

# Add labels and title
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Scatter Plot with Empty Circles")
```

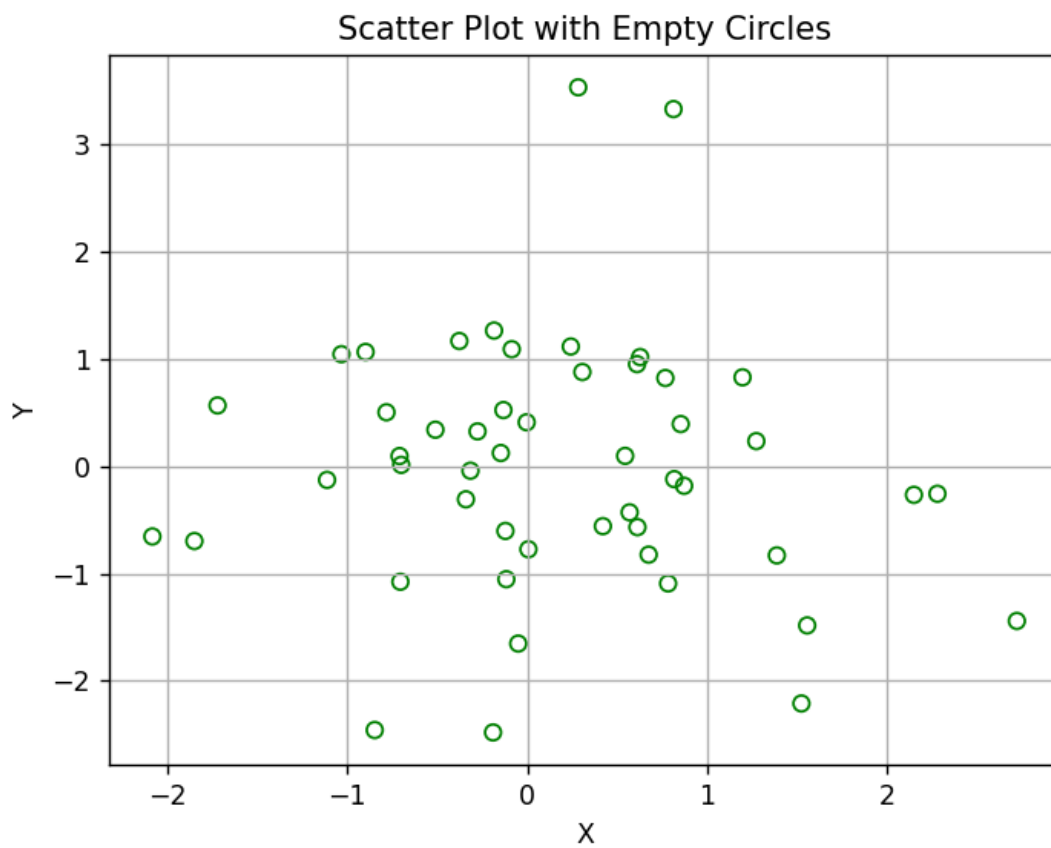
```
# Display grid
```

```
plt.grid(True)
```

```
# Show the plot
```

```
plt.show()
```

OUTPUT :



34. Write a Python program to draw a scatter plot using random distributions to generate balls of different sizes.

Code :

```
import numpy as np
import matplotlib.pyplot as plt

# Generate random X and Y coordinates
x = np.random.rand(20)
y = np.random.rand(20)

# Generate random sizes for the balls
sizes = np.random.randint(20, 300, 20)

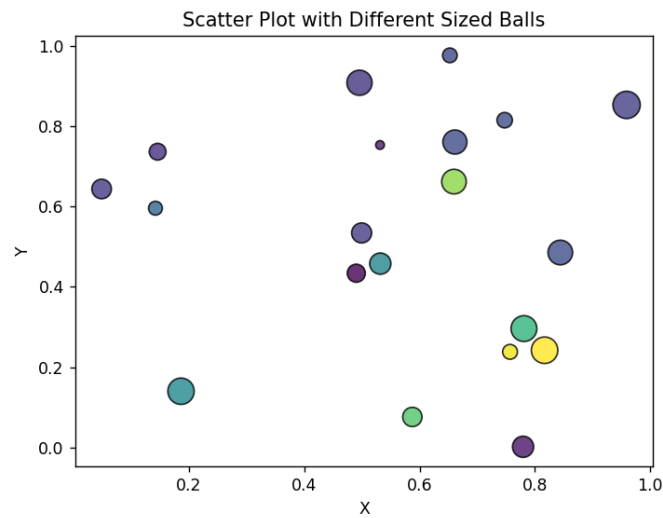
# Generate random colors
colors = np.random.rand(20)

# Create scatter plot
plt.scatter(
    x, y,
    s=sizes,
    c=colors,
    cmap='viridis',
    alpha=0.8,
    edgecolors='black')

# Labels and title
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Scatter Plot with Different Sized Balls")

plt.show()
```

OUTPUT :



35. Write a Python program to draw a scatter plot comparing two subject marks of Mathematics and Science. Use marks of 10 students.

Code :

```
import matplotlib.pyplot as plt

# Marks of 10 students
math_marks = [88, 92, 80, 89, 100, 80, 100, 80, 34, 90]
science_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30]

# Marks range
marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

# Scatter plot for Mathematics marks
plt.scatter(
    marks_range,
    math_marks,
```

```
    color='red',
    marker='.',
    label='Math marks'
)

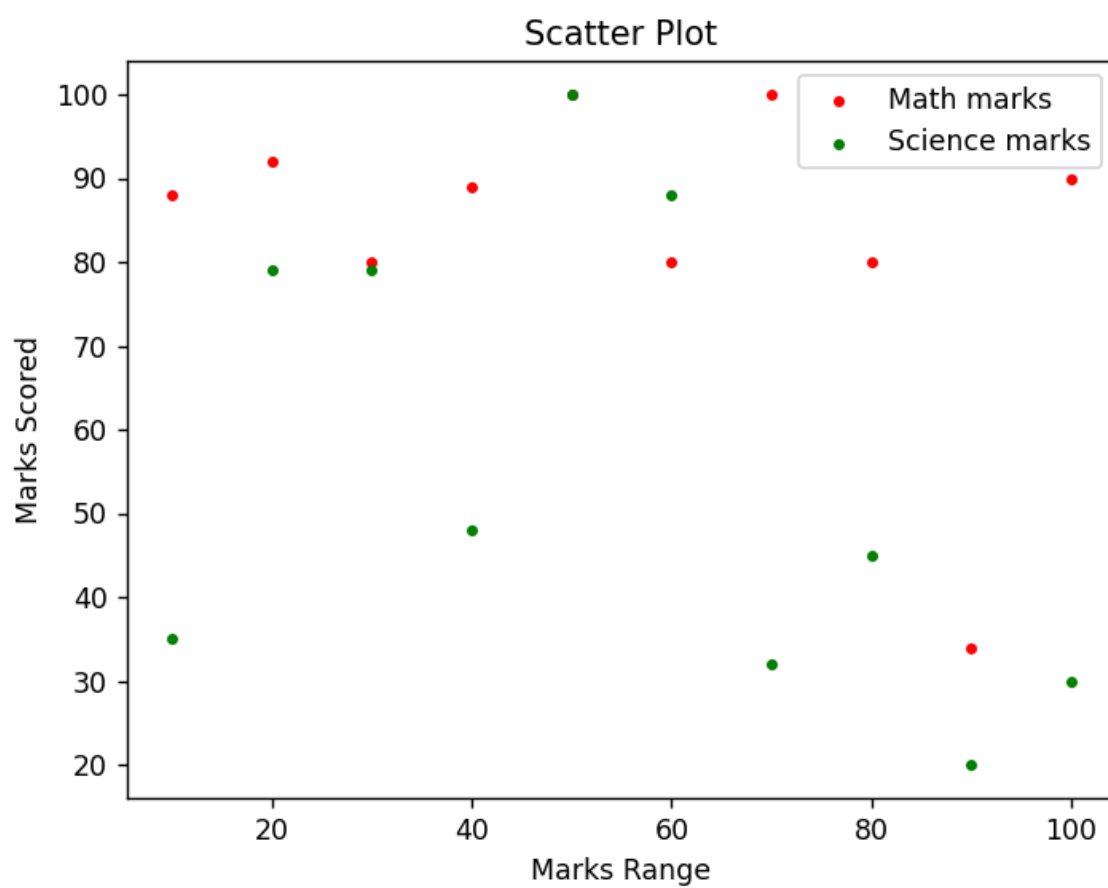
# Scatter plot for Science marks
plt.scatter(
    marks_range,
    science_marks,
    color='green',
    marker='.',
    label='Science marks'
)

# Labels and title
plt.xlabel("Marks Range")
plt.ylabel("Marks Scored")
plt.title("Scatter Plot")

# Display legend
plt.legend()

# Show the graph
plt.show()
```

OUTPUT :



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